

Supplementary Materials 1

To test whether conducting the experiment on personal computers with audio-visual (AV) stimuli presented in a web-browser gives sufficiently reliable audio-visual synchrony for single participant assessment of phase-lag effects, we tested our experimental protocol using both a MacBook and a Windows laptop (see Table 1 for hardware specifications). These are both more widely available than alternative OS platforms (Linux, Chromebook) and due to their prevalence in the Prolific population from which we recruited our participants. We tested only using the Chrome browser as it is the most commonly used browser in the general population, and a well-performing choice in the literature validating online stimulus presentation and response measurement (Bridges et al., 2020). We aimed to achieve a similar level of variance in audio-visual synchrony (5-6 ms SD) as Bridges et al. (2020) and hence, set a strict criterion of 5 ms SD lower.

We minimally modified the experiment to make recordings of AV synchrony as precise as possible. We created a single H.264- and AAC-(identical codecs and container, .mp4, to the clips used for the experiment) clip with alternating black and white frames each lasting 167-ms (i.e. 10-frames at the standard 60 Hz playback rate), and a short 1000 Hz tone that alternated with silence in synchrony with the black and white frames such that white frames coincided with the tones and black frames coincided with silent periods. As in the main experiment we constructed 192 trials each with a clip duration of 3000 ms. We used a photodiode (TSL257 High Sensitivity Light-to-Voltage Converter) to record the brightness on the screen. The output of the photodiode, along with the sound output was recorded by connecting each of these laptops to a USB sound interface (StarTech 7.1 USB Sound Card) with 3.5mm cable. The interface has stereo input that we could connect the photodiode as well. Using this setup, we could record millisecond precise changes in screen brightness (video) and sound output as two separate channels of a file lasting ~780 seconds (i.e. the duration of the experiment for only the encoding task) at a sample rate of 44100 Hz recorded using Audacity software (2.3.0, <https://www.audacityteam.org/>).

We analysed the resulting signals in MATLAB using scripts and recordings available on GitHub (<https://github.com/fserin/OptimalThetaPhase>). The sound signal was first bandpass filtered at 1000 Hz (with 900 and 1100 Hz lower and upper cut-offs respectively) using butterworth filter to remove noise. Then, the envelope of the audio signal was extracted using a Hilbert transformation. Thanks to our dual channel recording setup, a simple threshold method was able to extract the 192 audio-video trials per laptop recording. We then performed a cross-correlation analysis between video and sound signal in each trial. The highest correlation lag was saved as the delay between audio and video presentation for that specific trial.

We calculated means and standard deviations of these lags over the 192 trials in a single simulated experiment. While synchronous presentation (i.e. a mean 0ms delay) is desirable for most purposes, the standard deviation of the cross-trial lag is more critical for our proposed experiment. Our experiment depends on being able to detect the preferred audio-visual lag for memory encoding in single participants. We will not be able to distinguish variability caused by different participants' equipment, but to determine the preferred lag requires consistent and stable audio-visual timing over trials. Only if there is a small amount of variance in the audio-visual lag over trials (less than the 15.625ms difference between adjacent phase bins) can we determine the preferred phase lag for each participant.

The results of our experiment demonstrated noticeable differences between the two systems tested. The Windows laptop demonstrated an average delay of -85.75 ms (SD = 6.64 ms), with audio lagging behind video. MacBook displayed an average delay of 23.41 ms (SD = 4.79 ms). The calculated delays are plotted per trial and separately for each laptop in Figure 1. The results showed that while MacBook was able to meet our criteria of 5 ms variance between trials, the Windows laptop failed to achieve that. Although the difference between these two systems was not massive, and might be dependent on specific hardware details, given our criterion, we decided to collect data from only MacBook users.

Specifications	Laptop		
		MacBook	Windows Laptop

Model/Series	MacBook Pro (13-inch, 2017)	Dell Latitude 5410
Operating System	MacOS 13.7.8	Windows 11 (24H2)
Screen Resolution	2560x1600	1920x1080
Refresh Rate	60 Hz	60 Hz
Processor	Intel(R) Core(TM) i5-7267U CPU @ 3.10GHz 3.1 GHz	Intel(R) Core(TM) i5-10310U CPU @ 1.70GHz 2.21 GHz
Ram	8192 MB	8192 MB
Chrome Version	147.0.7727.56	148.0.7778.168

Table 1. The specifications of the two laptops that were tested.

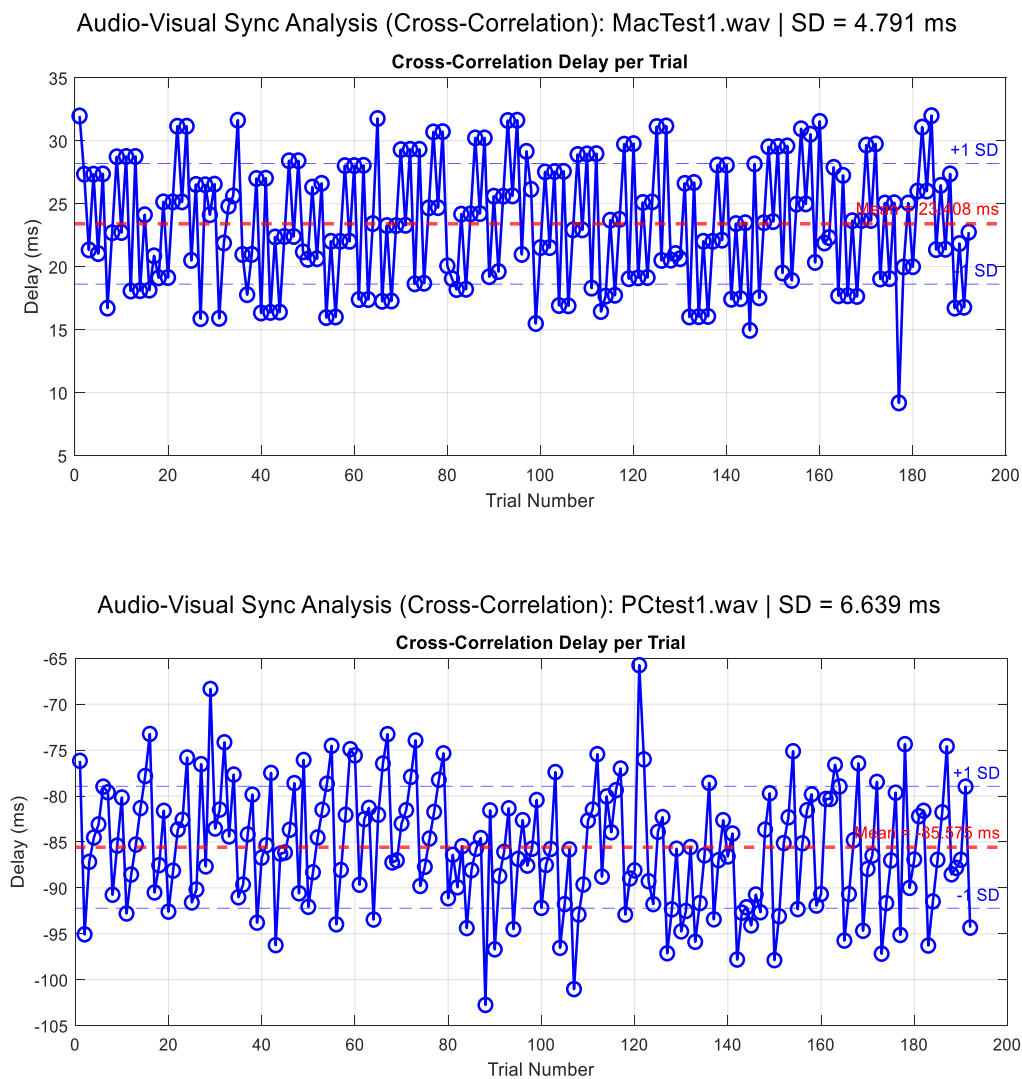


Figure 1. Calculated delays per trial for each laptop.

References

Bridges, D., Pitiot, A., MacAskill, M. R., & Peirce, J. W. (2020). The timing mega-study: Comparing a range of experiment generators, both lab-based and online. *PeerJ*, 8, e9414. <https://doi.org/10.7717/peerj.9414>